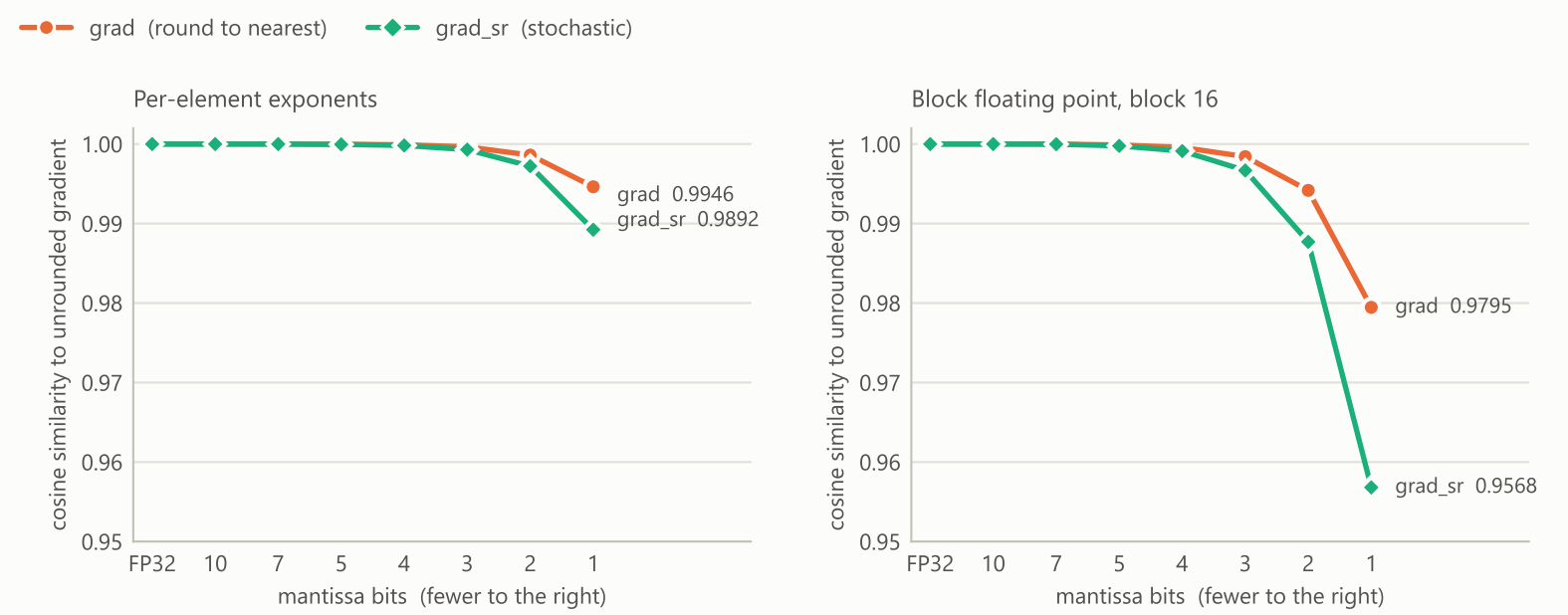
grad (round to nearest)
grad_sr (stochastic)
Per-element exponents
cosine similarity to unrounded gradient
1.00
0.99
0.98
0.97
0.96
0.95
grad 0.9946
grad_sr 0.9892
FP32
10
7
5
4
3
2
1
mantissa bits (fewer to the right)
Block floating point, block 16
cosine similarity to unrounded gradient
1.00
0.99
0.98
0.97
0.96
0.95
grad 0.9795
grad_sr 0.9568
FP32
10
7
5
4
3
2
1
mantissa bits (fewer to the right)